

Year 7 • Chapter 5B • Equations: answers (teacher copy)

5.10 Guess, check and improve (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: yes
- Ex 1b: $m = 4$
- Ex 2a: $k = 0$
- Ex 2b: $d = 18$
- Ex 3a: $a = 9$
- Ex 3b: $x = 0.5$

Easy (drill)

- | | | |
|---------|--------------|--------------|
| 1. no | 11. yes | 21. $k = 3$ |
| 2. yes | 12. yes | 22. $m = 14$ |
| 3. no | 13. $p = 10$ | 23. $a = 14$ |
| 4. yes | 14. $b = 10$ | 24. $b = 10$ |
| 5. yes | 15. $k = 16$ | 25. $m = 12$ |
| 6. yes | 16. $m = 1$ | 26. $w = 10$ |
| 7. no | 17. $x = 14$ | 27. $k = 15$ |
| 8. no | 18. $m = 14$ | 28. $a = 8$ |
| 9. no | 19. $y = 4$ | |
| 10. yes | 20. $t = 9$ | |

Medium (drill)

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|-------------|-------------|--------------|
| 1. $n = 15$ | 5. $w = 12$ | 9. $w = 10$ |
| 2. $a = 7$ | 6. $b = 10$ | 10. $n = 5$ |
| 3. $b = 14$ | 7. $a = 12$ | 11. $w = 5$ |
| 4. $y = 6$ | 8. $t = 4$ | 12. $k = 11$ |

Challenging

- $x = 0.5$
- $n = 12$ (or -12)
- $x = 5$ (or -6)

Extension

- $x \approx 3.7$
- $x \approx 6.6$

Exit ticket

- Level 1: yes
- Level 2: $x = 7$
- Level 3: $y = 20$

5.11 Solving one-step equations (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $w = 35$
- Ex 1b: $e = 12$
- Ex 2a: $p = 3$
- Ex 2b: $x = 12$
- Ex 3a: $y = -7$
- Ex 3b: $k = -2.5$

Easy (drill)

- | | | |
|--------------|--------------|--------------|
| 1. $k = 25$ | 12. $n = 10$ | 23. $k = 20$ |
| 2. $w = 17$ | 13. $t = 23$ | 24. $b = 6$ |
| 3. $w = 9$ | 14. $n = 29$ | 25. $p = 48$ |
| 4. $b = 26$ | 15. $n = 32$ | 26. $a = 10$ |
| 5. $b = 39$ | 16. $b = 3$ | 27. $p = 35$ |
| 6. $w = 13$ | 17. $k = 11$ | 28. $a = 56$ |
| 7. $w = 41$ | 18. $t = 5$ | 29. $n = 5$ |
| 8. $t = 11$ | 19. $p = 18$ | 30. $x = 56$ |
| 9. $w = 25$ | 20. $y = 9$ | 31. $x = 24$ |
| 10. $k = 22$ | 21. $a = 18$ | 32. $k = 11$ |
| 11. $t = 14$ | 22. $n = 33$ | |

Medium (drill)

- | | | |
|--------------|--------------------------------|--------------------------------|
| 1. $n = 6$ | 11. $m = -15$ | 19. $x = \frac{16}{6} = 2.667$ |
| 2. $t = -8$ | 12. $p = 1$ | 20. $x = \frac{1}{3} = 0.333$ |
| 3. $y = -3$ | 13. $m = 3$ | 21. $y = \frac{16}{5} = 3.2$ |
| 4. $x = 8$ | 14. $n = -14$ | 22. $y = \frac{19}{2} = 9.5$ |
| 5. $w = -9$ | 15. $y = 8$ | 23. $w = \frac{7}{3} = 2.333$ |
| 6. $m = -11$ | 16. $x = 12$ | 24. $a = \frac{35}{2} = 17.5$ |
| 7. $n = -4$ | 17. $w = \frac{3}{10} = 0.3$ | |
| 8. $t = 8$ | 18. $y = \frac{20}{7} = 2.857$ | |
| 9. $t = 11$ | | |
| 10. $y = 1$ | | |

Challenging

- $n + 19 = 45$, $n = 26$
- $7n = 84$, $n = 12$
- $n/4 = 11$, $n = 44$

Extension

- $n = 4$
- $a = 15$

Exit ticket

- Level 1: $n = 22$
- Level 2: $x = 7$
- Level 3: $t = -24$

5.12 Solving two-step equations (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: $t = 2$
- Ex 1b: $y = 3$
- Ex 2a: $m = 68$
- Ex 2b: $x = 9$
- Ex 3a: $m = 10$
- Ex 3b: $x = -4$

Easy (drill)

- | | | |
|--------------|--------------|--------------|
| 1. $m = 8$ | 12. $y = 4$ | 23. $x = 18$ |
| 2. $w = 4$ | 13. $n = 6$ | 24. $t = 3$ |
| 3. $b = 11$ | 14. $t = 3$ | 25. $b = 18$ |
| 4. $a = 5$ | 15. $x = 10$ | 26. $k = 24$ |
| 5. $k = 10$ | 16. $k = 9$ | 27. $w = 9$ |
| 6. $k = 8$ | 17. $k = 24$ | 28. $k = 18$ |
| 7. $b = 6$ | 18. $y = 10$ | 29. $p = 10$ |
| 8. $n = 1$ | 19. $a = 42$ | 30. $p = 3$ |
| 9. $y = 6$ | 20. $m = 24$ | 31. $n = 16$ |
| 10. $x = 10$ | 21. $y = 30$ | 32. $y = 48$ |
| 11. $p = 11$ | 22. $a = 16$ | |

Medium (drill)

- | | | |
|-------------|--------------|--------------|
| 1. $n = -4$ | 9. $t = -1$ | 17. $w = 1$ |
| 2. $y = -4$ | 10. $y = -4$ | 18. $y = 3$ |
| 3. $p = 7$ | 11. $k = -8$ | 19. $a = 12$ |
| 4. $t = -8$ | 12. $w = 2$ | 20. $m = 8$ |
| 5. $p = -1$ | 13. $b = -9$ | 21. $x = 6$ |
| 6. $b = -3$ | 14. $w = -3$ | 22. $t = 11$ |
| 7. $k = 6$ | 15. $w = -1$ | |
| 8. $t = 9$ | 16. $a = -9$ | |

Challenging

- $x = 12$
- $y = 9.5$
- $t = 4$

Extension

- $a = 4$

2. $x = 4$

Exit ticket

1. Level 1: $x = 6$
2. Level 2: $n = 18$
3. Level 3: $k = 9$

5.13 Equations and problem solving

(booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: C: $17N = 1003$
2. Ex 1b: $N = 59$
3. Ex 2a: $5N = 180$, $N = 36$ tickets
4. Ex 2b: $5N - 180 = 2100$, $N = 456$ tickets
5. Ex 3a: $4s = 52$, $s = 13$ cm
6. Ex 3b: $n + n + 1 + n + 2 = 72$: 23, 24, 25

Easy (drill)

- | | | |
|--------------------------------|--------------------------------|--|
| 1. $2n + 8 = 36$, $n =$
14 | 5. $2n + 9 = 41$, $n =$
16 | 10. $2n + 6 = 16$
11. $n \div 6 = 20$ |
| 2. $7n = 98$, $n = 14$ | 6. $8n = 152$, $n = 19$ | 12. $2n + 4 = 44$ |
| 3. $2n = 30$, $n = 15$ | 7. $2n + 2 = 36$ | 13. $3n - 9 = 51$ |
| 4. $n + 2 = 18$, $n =$
16 | 8. $2n + 5 = 27$ | 14. $n - 8 = 11$ |
| | 9. $4n = 12$ | |

Medium (drill)

1. $10n = 160$, 16 pencils
2. $4s = 36$, 9 cm
3. $4s = 16$, 4 cm
4. $9n = 81$, 9 pencils
5. $14n = 168$, 12 pencils
6. $4s = 28$, 7 cm
7. $49 + 32 + x = 180$, $x = 99$
8. $50 + 34 + x = 180$, $x = 96$
9. $42 + 40 + x = 180$, $x = 98$
10. $37 + 58 + x = 180$, $x = 85$

Challenging

1. 23, 24, 25
2. length 16 m, width 8 m
3. 456 tickets

Extension

1. 7.5 weeks
2. 32 and 18

Exit ticket

1. Level 1: $n + 14 = 31$, $n = 17$
2. Level 2: $4b = 62$, $b = \$15.50$
3. Level 3: $2n + 9 = 41$, $n = 16$